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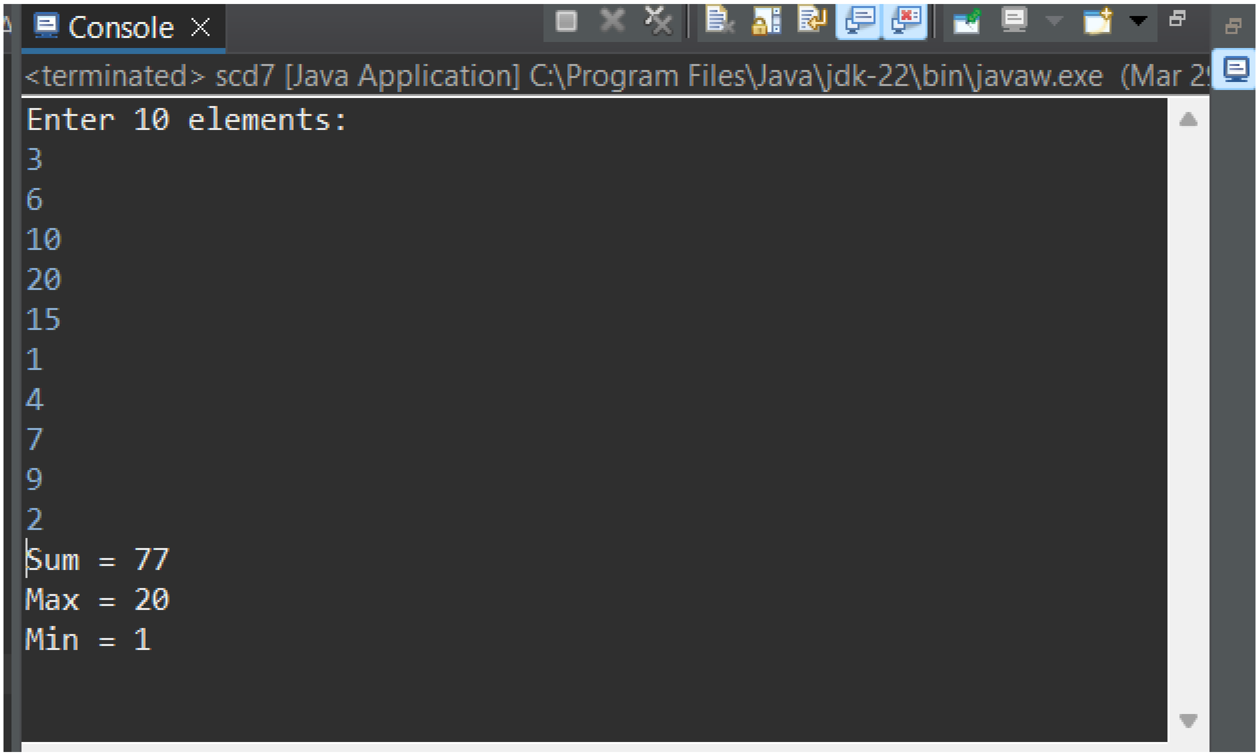
ROLL NO: 23-SE-64

TASK 1:

```
import java.util.*;

public class ArrayTask1 {
   public static void main(String[] args) {
      int[] arr = new int[10];
      Scanner sc = new Scanner(System.in);

      System.out.println("Enter 10 elements:");

      // Input
      for (int i = 0; i < 10; i++) {
         arr[i] = sc.nextInt();
      }

      int sum = 0;
      int max = arr[0];
      int min = arr[0];

      // Processing
      for (int i = 0; i < 10; i++) {
         sum += arr[i];

         if (arr[i] > max)
            max = arr[i];

         if (arr[i] < min)
            min = arr[i];
      }

      // Output
      System.out.println("Sum = " + sum);
      System.out.println("Max = " + max);
      System.out.println("Min = " + min);
   }
}
```

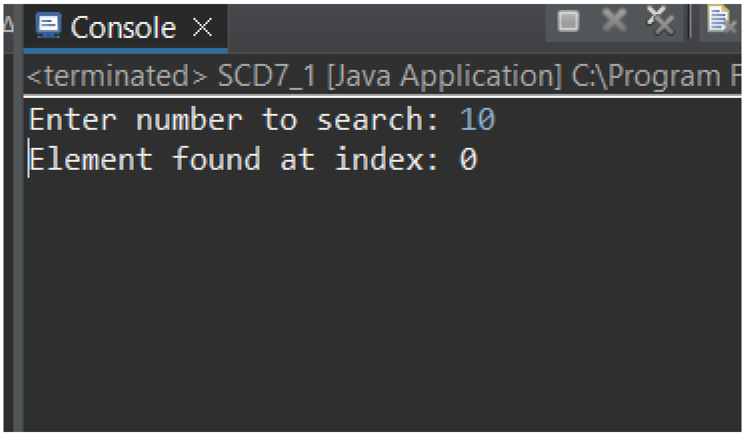
OUTPUT:



TASK 2:

```
import java.util.*;

public class ArraySearch {
  public static void main(String[] args) {
    int[] arr = {10, 20, 30, 40, 50};
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number to search: ");
    int key = sc.nextInt();

    boolean found = false;

    for (int i = 0; i < arr.length; i++) {
      if (arr[i] == key) {
        System.out.println("Element found at index: " + i);
        found = true;
        break;
      }
    }

    if (!found) {
      System.out.println("Element not found");
    }
  }
}
```

OUTPUT:

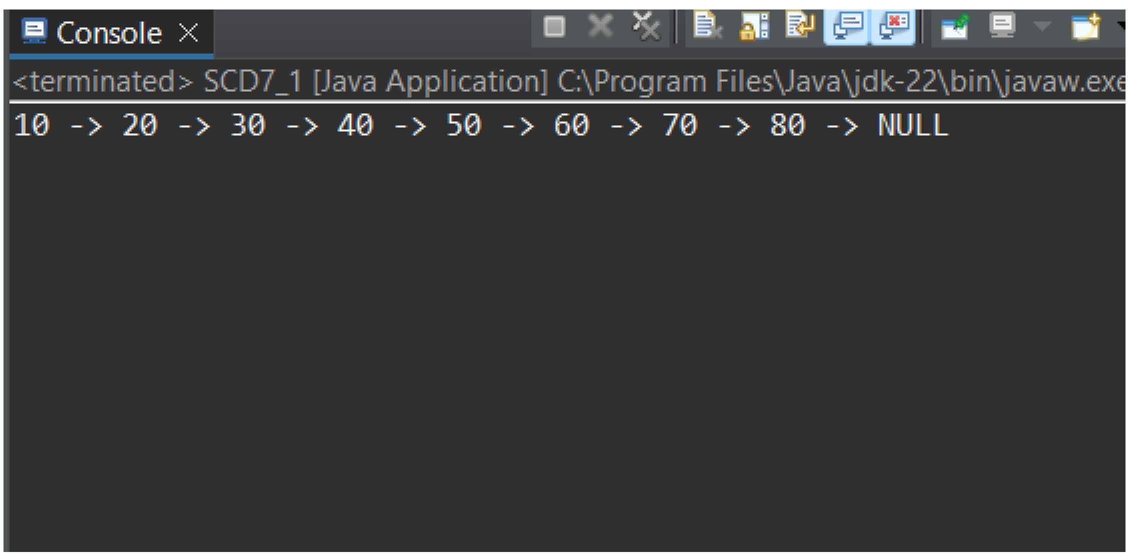


TASK 3:

```
class Node {
  int data;
  Node next;

  Node(int data) {
    this.data = data;
    this.next = null;
  }
}

public class LinkedListTask3 {
  Node head;

  // Insert at end
  public void insert(int data) {
    Node newNode = new Node(data);

    if (head == null) {
      head = newNode;
      return;
    }

    Node temp = head;
    while (temp.next != null) {
      temp = temp.next;
    }

    temp.next = newNode;
  }

  // Display list
  public void display() {
```

```
    Node temp = head;
    while (temp != null) {
      System.out.print(temp.data + " -> ");
      temp = temp.next;
    }
    System.out.println("NULL");
  }

  public static void main(String[] args) {
    LinkedListTask3 list = new LinkedListTask3();

    // Insert 8 nodes
    for (int i = 1; i <= 8; i++) {
      list.insert(i * 10);
    }

    list.display();
  }
}
```

OUTPUT:

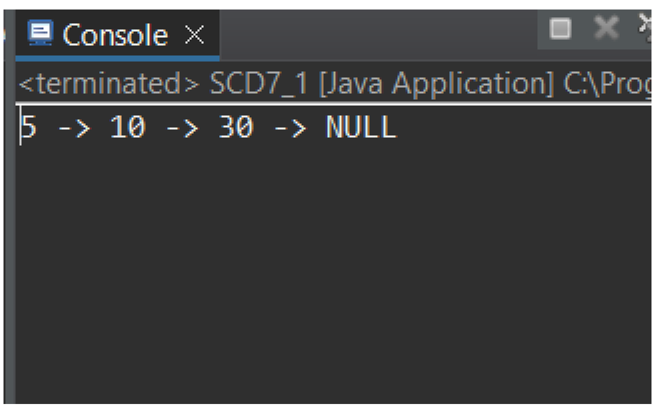


TASK 4:

```
public class LinkedListTask4 {
    Node head;

    // Insert at beginning
    public void insertBeginning(int data) {
        Node newNode = new Node(data);
        newNode.next = head;
        head = newNode;
    }

    // Insert at end
    public void insertEnd(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    // Delete a node
    public void delete(int key) {
        Node temp = head, prev = null;

        if (temp != null && temp.data == key) {
            head = temp.next;
            return;
        }

        while (temp != null && temp.data != key) {
            prev = temp;
            temp = temp.next;
        }

        if (temp == null) return;

        prev.next = temp.next;
    }

    // Display
```

```
  public void display() {
    Node temp = head;
    while (temp != null) {
      System.out.print(temp.data + " -> ");
      temp = temp.next;
    }
    System.out.println("NULL");
  }

  public static void main(String[] args) {
    LinkedListTask4 list = new LinkedListTask4();

    list.insertEnd(10);
    list.insertEnd(20);
    list.insertEnd(30);

    list.insertBeginning(5); // insert at start
    list.delete(20); // delete node

    list.display();
  }
}
```

OUTPUT:

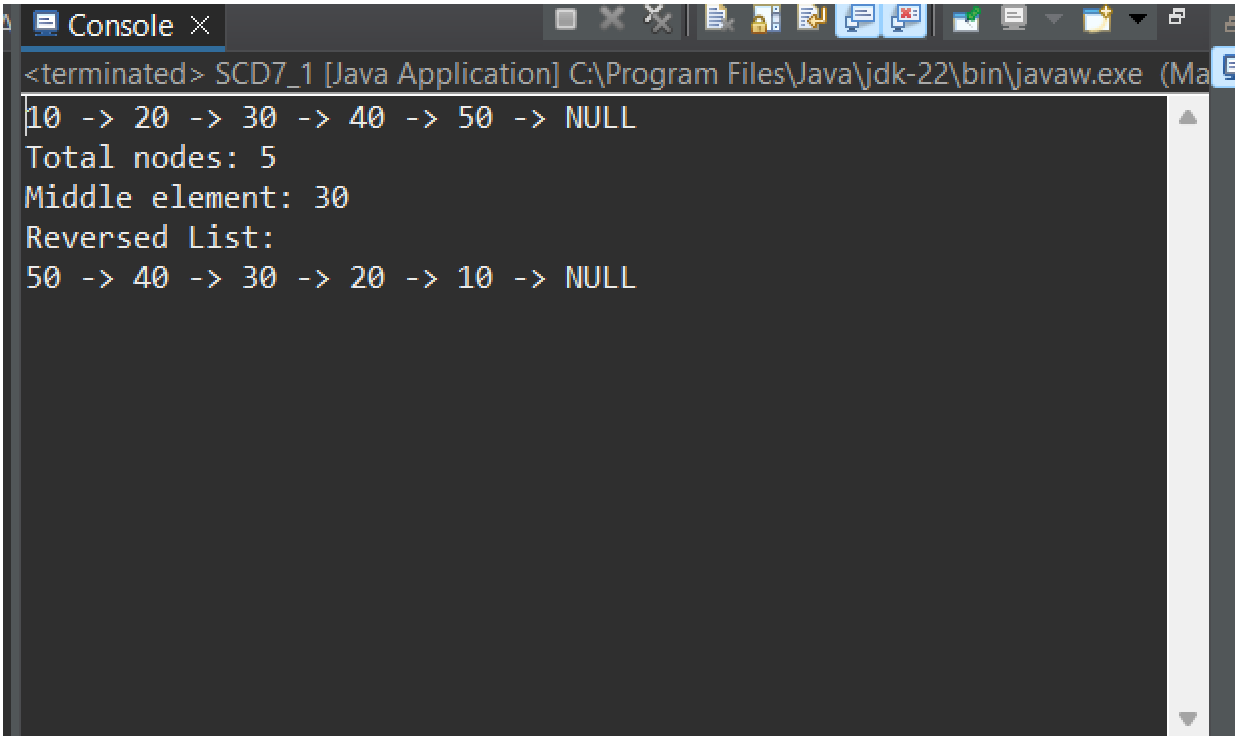


TASK 5:

```
public class LinkedListTask5 {
    Node head;

    public void insert(int data) {
        Node newNode = new Node(data);

        if (head == null) {
            head = newNode;
            return;
        }

        Node temp = head;
        while (temp.next != null) {
            temp = temp.next;
        }

        temp.next = newNode;
    }

    // Count nodes
    public int count() {
        int count = 0;
        Node temp = head;

        while (temp != null) {
            count++;
            temp = temp.next;
        }
        return count;
    }

    // Find middle
    public void findMiddle() {
        Node slow = head;
        Node fast = head;

        while (fast != null && fast.next != null) {
            slow = slow.next;
            fast = fast.next.next;
        }

        System.out.println("Middle element: " + slow.data);
    }

    // Reverse list
    public void reverse() {
        Node prev = null;
```

```
    Node current = head;
    Node next;

    while (current != null) {
      next = current.next;
      current.next = prev;
      prev = current;
      current = next;
    }

    head = prev;
  }

  // Display
  public void display() {
    Node temp = head;
    while (temp != null) {
      System.out.print(temp.data + " -> ");
      temp = temp.next;
    }
    System.out.println("NULL");
  }

  public static void main(String[] args) {
    LinkedListTask5 list = new LinkedListTask5();

    list.insert(10);
    list.insert(20);
    list.insert(30);
    list.insert(40);
    list.insert(50);

    list.display();

    System.out.println("Total nodes: " + list.count());

    list.findMiddle();

    list.reverse();
    System.out.println("Reversed List:");
    list.display();
  }
}
```

OUTPUT:


Console
<terminated> SCD7_1 [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (Ma
10 -> 20 -> 30 -> 40 -> 50 -> NULL
Total nodes: 5
Middle element: 30
Reversed List:
50 -> 40 -> 30 -> 20 -> 10 -> NULL